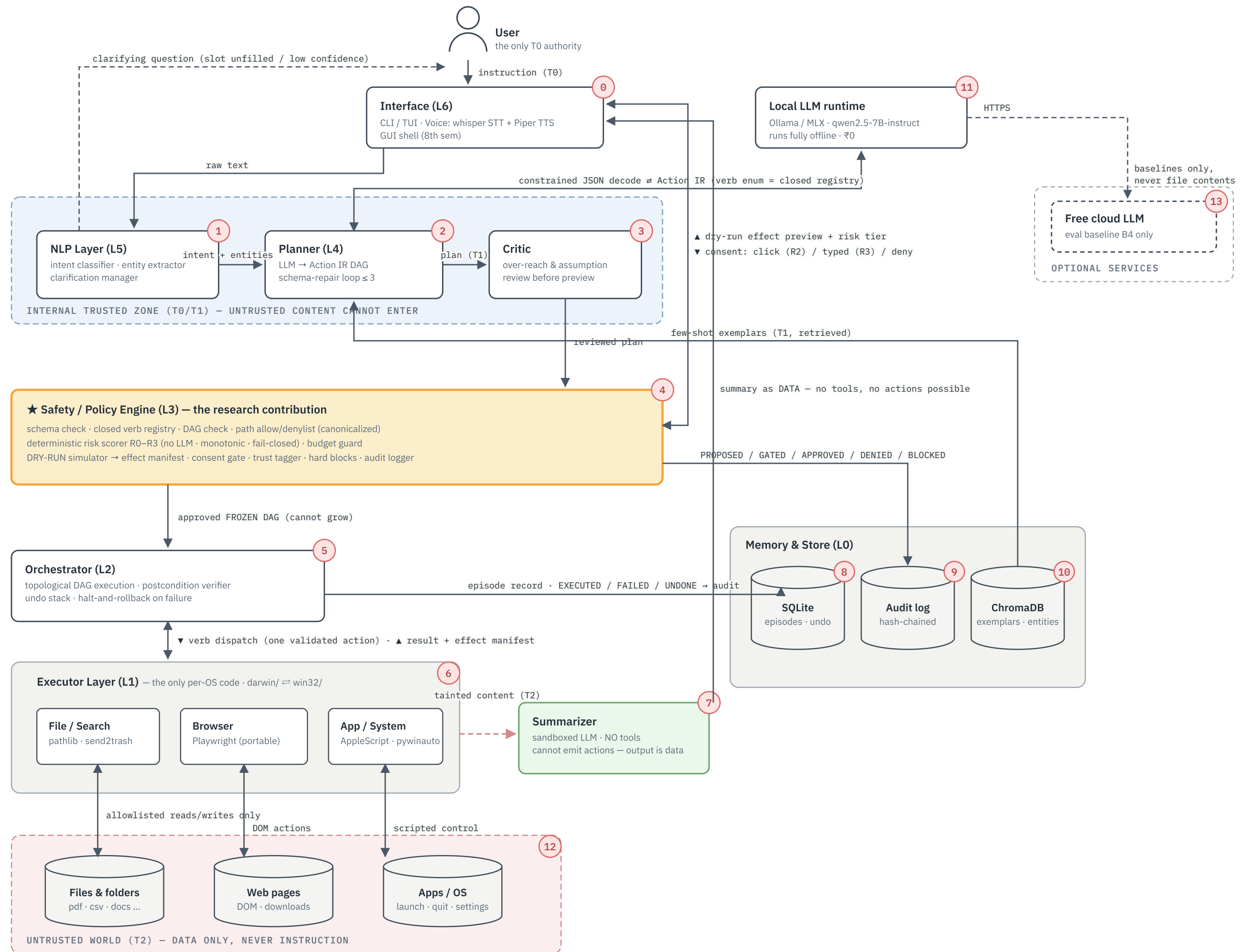


MAESTRO Architecture Map

Every component is numbered ①–⑬ and every line is labeled with what actually crosses it — because in this system, what crosses each boundary *is* the safety argument. Amber marks the research contribution; solid arrows carry trusted/typed data, dashed red carries untrusted content.



 safety layer — the research contribution  trusted zone (T0/T1)  untrusted (T2) — data only  sandboxed, tool-less  optional / dashed = not required  component number ①–⑬

Component index

#	COMPONENT	ROLE	STATUS (AUG 2026)
0	Interface (L6)	CLI/TUI today; voice (local whisper/Piper) Week 8; GUI in 8th sem	CLI working
1	NLP layer (L5)	Intent + entities + clarification; refusal is a trained class	Spec done; build W8+
2	Planner (L4)	Local LLM → Action IR DAG via constrained JSON decoding	Working (v0.2)
3	Critic	Reviews plans for over-reach before the user sees a preview	Spec done; 8th sem
4	Safety / policy engine (L3)	Deterministic gatekeeper: registry, paths, risk R0–R3, dry-run, consent, audit	Working + tested
5	Orchestrator (L2)	Executes the frozen DAG; verifies postconditions; rollback	Working + tested
6	Executors (L1)	Plain code, one per verb; the only per-OS layer (darwin/win32)	7 file verbs (darwin)
7	Summarizer	Only LLM allowed to read untrusted content; has no tools — the injection firewall	Spec done; 8th sem
8	SQLite store	Episodic memory + undo stack — raw material of the learning loop	Working (v0.2)
9	Audit log	Append-only, hash-chained: sha256(prev_hash row); tampering breaks the chain	Working + tested
10	ChromaDB	Workflow exemplars retrieved as few-shot context; learned entities	Week 10
11	Local LLM runtime	Ollama/MLX, qwen2.5-7B default; fine-tuned 3B swaps in via MAESTRO_MODEL	Working
12	Untrusted world	Files, web, apps — everything read from here is T2: data, never instruction	—
13	Cloud LLM (optional)	Free-tier frontier model, used only as evaluation baseline B4/M5	—